Systematicity between Forms and Meanings across Languages Supports Efficient Communication

语言间形式与意义的系统性支持高效沟通

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Abstract

摘要

Languages vary widely in how meanings map to word forms. These mappings have been found to support efficient communication; however, this theory does not account for systematic relations within word forms. We examine how a restricted set of grammatical meanings (e.g. person, number) are expressed on verbs and pronouns across typologically diverse languages. Consistent with prior work, we find that verb and pronoun forms are shaped by competing communicative pressures for simplicity (minimizing the inventory of grammatical distinctions) and accuracy (enabling recovery of intended meanings). Crucially, our proposed model uses a novel measure of complexity (inverse of simplicity) based on the learnability of meaning-to-form mappings. This innovation captures fine-grained regularities in linguistic form, allowing better discrimination between attested and unattested systems, and establishes a new connection from efficient communication theory to systematicity in natural language.

语言在意义如何映射到词形方面差异很大。这些映射已被发现支持有效的交流；然而，这一理论无法解释词形内部的系统性关系。我们考察了在类型学上差异很大的语言中，如何通过有限的一组语法意义（例如人称、数）在动词和代词中表达。与以往的研究一致，我们发现动词和代词的形式受到简化（最小化语法区别的库存）和准确性（使意图意义的恢复成为可能）这两种竞争性交流压力的影响。关键的是，我们提出的模型使用了一种基于意义到词形映射可学习性的复杂度的新度量（即简化度的倒数）。这一创新捕捉了语言形式中精细的规律性，使我们能够更好地区分已证实和未证实的系统，并在有效交流理论与自然语言中的系统性之间建立了新的联系。

1 Introduction

1 引言

Languages express a vast array of meanings with a limited set of forms. A fundamental goal of natural language research is to understand how these forms map onto meanings, and why this mapping differs across languages. For example, standard English uses the pronoun you for any second-person addressee, while other languages use distinct forms based on features like gender or number (e.g., sen “you (singular)” vs. siz “you (plural)” in Turkish; Figure 1). What drives this cross-linguistic variation in form-meaning mappings?

语言用有限的形式表达大量的含义。自然语言研究的一个基本目标是理解这些形式如何映射到含义，以及为什么这种映射在不同语言中存在差异。例如，标准英语使用代词 “you” 来指代任何第二人称的对话对象，而其他语言则根据诸如性别或人称等特征使用不同的形式（例如，土耳其语中的 “sen” 表示 “你（单数）”，“siz” 表示 “你（复数）”；图 1)。是什么驱动了形式与含义映射的跨语言差异?

We posit that this variation arises because languages evolve under two competing communicative pressures. On one hand, a general bias toward simplicity encourages languages to minimize formal distinctions. This criterion favors the English pronoun system, with only one second-person form, as less cognitively demanding than Turkish, which

我们认为这种差异的出现是因为语言在两种相互竞争的交流压力下演变。一方面，一种普遍的简洁偏好促使语言尽量减少形式上的差异。这一标准倾向于认为英语代词系统，只有一个人称代词形式，比土耳其语更少认知负担，因为后者

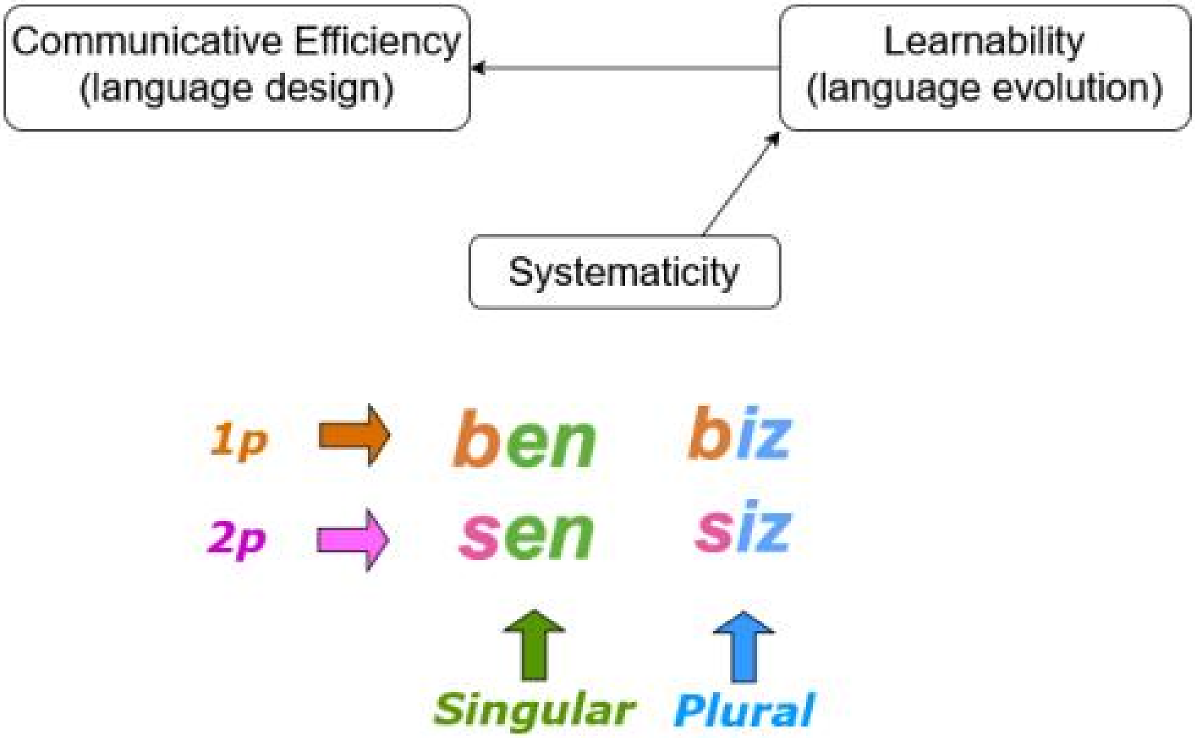

Figure 1: images/fb504e2a856501920dcb99e3d1d28a1ec047f8ddc6cd55d65fd297bc8b6b3777.jpg

Figure 1: Turkish pronouns show systematic form-meaning mappings: person is consistently marked by prefixes (e.g., s- for second person), number by suffixes. Language evolution research demonstrates that such systematicity supports learnability. Our model connects these findings, proposing that learnable, systematic mappings contribute to communicative efficiency.

图 1：土耳其语代词表现出系统性的形式-意义映射：人称由前缀一致标记（例如，s-表示第二人称），数由后缀表示。语言演化研究显示，这种系统性支持学习性。我们的模型将这些发现联系起来，提出可学习的、系统性的映射有助于交流效率。

requires the speaker to distinguish addresses by number. On the other hand, a competing bias toward accuracy encourages languages to maintain formal distinctions, enabling precise decoding of meaning from form. For instance, a Turkish listener can instantly understand whether the speaker intends to address an individual (sen) or a group (siz), while an English listener must infer between these two meanings of the word you. English and Turkish pronoun systems illustrate two different design solutions to the trade-off between simplicity and accuracy (Zaslavsky et al., 2021b).

要求说话者通过数字来区分地址。另一方面，一种追求准确性的竞争性偏见促使语言保持正式的区分，从而能够从形式中精确地解码意义。例如，土耳其语的听者可以立即理解说话者是想称呼个人（sen）还是群体（siz），而英语的听者则必须在 “you” 这个词的两种含义之间进行推断。英语和土耳其语的人称系统展示了两种不同的设计解决方案，用于在简单性和准确性之间进行权衡（Zaslavsky 等，2021b）。

This trade-off has been explored through two distinct research traditions. An influential line of work on efficient communication (e.g. Kemp et al., 2018; Gibson et al., 2019; Hahn et al., 2020) analyzes how natural languages balance simplicity and accuracy in their lexicons. This research finds that lexicons achieve near-optimal simplicity-accuracy trade-offs across semantic domains including color (Zaslavsky et al., 2018), kinship (Kemp and Regier, 2012), and numerals (Xu et al., 2020). In parallel, research on language evolution (e.g. Smith et al., 2017; Culbertson and Schuler, 2019) investigates

这种权衡通过两种不同的研究传统进行了探讨。一项有影响力的研究工作（例如 Kemp 等，2018；Gibson 等，2019；Hahn 等，2020）分析了自然语言如何在其词汇中平衡简洁性和准确性。这项研究发现，词汇在包括颜色（Zaslavsky 等，2018）、亲属关系（Kemp 和 Regier，2012）和数字（Xu 等，2020）等语义领域中实现了接近最优的简洁性-准确性权衡。同时，关于语言演化的研究（例如 Smith 等，2017；Culbertson 和 Schuler，2019）则探讨了……

how artificial languages evolve in laboratory settings. Through experimental manipulation, this work shows that interacting biases toward simplicity and accuracy drive the emergence of core “design features” of natural language (Hockett, 1960; Christiansen and Chater, 2008; Smith, 2022).

人工语言在实验室环境中的演变。通过实验操纵，本研究展示了对简单性和准确性的相互偏见如何驱动自然语言核心 “设计特征” 的出现（Hockett, 1960; Christiansen 和 Chater, 2008; Smith, 2022）。

One such core feature is systematicity, [1] in which discrete parts of linguistic forms reliably indicate specific meanings; consider the Turkish pronouns in Figure 1, where prefixes indicate person (e.g. second person pronouns begin with s-) and suffixes indicate number. Systematic mappings emerge in artificial languages as a design solution to the competing pressures of simplicity and accuracy (Smith et al., 2013; Kirby et al., 2015; Smith, 2022; Smith and Culbertson, 2025): decomposing complex expressions into parts yields a smaller, simpler lexicon, while recombining these parts enables fine-grained distinctions and accurate decoding. The language evolution literature thus identifies systematic form-meaning mappings as a core mechanism of efficient communication. However, this insight is not yet reflected in research on communicative efficiency in natural languages, which lacks a general framework for modeling forms’ internal structure.

一个这样的核心特征是系统性，[1] 即语言形式的离散部分可靠地指示特定含义；例如图 1 中的土耳其代词，其中前缀表示人称（例如第二人称代词以 “s-” 开头），后缀表示数。系统性的映射在人工语言中作为一种设计解决方案出现，以应对简洁性和准确性的竞争压力（Smith 等，2013；Kirby 等，2015；Smith，2022；Smith 和 Culbertson，2025）：将复杂表达分解为部分可以产生更小、更简单的词汇表，而重新组合这些部分则能够实现精细的区分和准确的解码。因此，语言演化文献将系统性的形式-含义映射识别为高效沟通的核心机制。然而，这一见解尚未体现在自然语言中沟通效率的研究中，因为这些研究缺乏一个用于建模形式内部结构的一般框架。

We propose a unified information-theoretic framework to integrate these two research traditions and assess how systematic form-meaning mapping supports efficiency in natural language. Our key contribution is a novel complexity measure based on the learnability of meaning-to-form mappings. This measure captures systematicity in linguistic forms within the theoretical framework of efficient communication.

我们提出一个统一的信息论框架，以整合这两种研究传统，并评估系统性的形式-意义映射如何支持自然语言中的效率。我们的主要贡献是一个基于意义到形式映射可学习性的新复杂性度量。这一度量在高效交流的理论框架内捕捉了语言形式中的系统性。

We evaluate our framework in two domains: verbs and pronouns. Verbs offer rich variation in how structured meanings map to forms across languages, and in many languages exhibit overtly systematic form-meaning covariation through inflectional morphology (Haspelmath and Sims, 2010). Pronouns provide a natural comparison case to an earlier analysis under the well-known Information Bottleneck model (Zaslavsky et al., 2021b), allowing us to directly compare approaches.

我们在两个领域评估我们的框架：动词和代词。动词在不同语言中，结构意义如何映射到形式方面表现出丰富的变化，并且在许多语言中，通过屈折形态表现出明显系统性的形式-意义共变（Haspelmath 和 Sims，2010）。代词为与众所周知的信息瓶颈模型（Zaslavsky 等，2021b）下的早期分析进行自然比较提供了案例，使我们能够直接比较不同的方法。

We find that our framework successfully represents structured form-meaning mappings in both domains, and consistently discriminates attested from unattested systems. Crucially, our comparison finds that explicitly modeling systematicity

我们发现我们的框架在两个领域中都能成功地表示结构化的形式-意义映射，并且能够一致地区分已证实的和未证实的系统。关键的是，我们的比较发现，显式地建模系统性

		singular (sg)	dual (du)	plural (pl)
1	m	?a-12u3-u	na-12u3-u	na-12u3-u
f	?a-12u3-u	na-12u3-u	na-12u3-u	
2	m	ta-12u3-u	ta-12u3-āni	ta-12u3-ūna
f	ta-12u3-īna	ta-12u3-āni	ta-12u3-na	
3	m	ya-12u3-u	ya-12u3-āni	ya-12u3-ūna
f	ta-12u3-u	ta-12u3-āni	ya-12u3-na	

		单数（sg）	双（du）	复数 (pl)
1	m	?a-12u3-u	NA-12U3-U	NA-12U3-U
f	?a-12u3-u	na-12u3-u	na-12u3-u	
2	m	TA-12U3-U	TA-12U3-āni	TA-12U3-ūna
f	ta-12u3-īna	TA-12U3-ĀNI	ta-12u3-na	
3	m	ya-12u3-u	ya-12u3-āni	ya-12u3-ūna
f	TA-12U3-U	TA-12U3-āni	ya-12u3-na	

Table 1: Basic imperfective paradigm of a stem I Classical Arabic verb for feature values gender {masculine, feminine}, person {1, 2, 3}, and number {sg, du, pl}. Numbers (e.g. 12u3) represent root consonants.

表 1: 一个以元音 I 为词干的古典阿拉伯语动词的不完全体基本形式，其特征值包括性别 {阳性，阴性}，人称 {1，2，3}，以及数 {单数，双数，复数}。数字（例如 12u3）表示词根辅音。

yields a more precise account of efficiency in language, better capturing the fine-grained regularities that characterize natural linguistic systems.

提供了一种更精确的语言效率描述，更好地捕捉自然语言系统所具有的细致的规律性。

2 Background

2 背景

2.1 Paradigms and Syncretism

2.1 模式与融合

A paradigm comprises systematic relations between word forms which are structured by grammatical categories such as person, gender, number, tense, or case. These relations can hold between distinct words, such as the Turkish pronouns shown in Figure 1, or parts of words, such as the prefixes and suffixes shown in Table 1. Specific feature combinations are realized by the form in the corresponding cell of the paradigm table (e.g. first person singular □ ?a-).

一个范式包括词形之间的系统性关系，这些关系由诸如人称、性别、数、时态或格等语法范畴所结构化。这些关系可以存在于不同的词之间，例如图 1 中所示的土耳其语代词，或者词的组成部分之间，例如表 1 中所示的前缀和后缀。特定的特征组合通过范式表中相应单元格的词形来实现（例如第一人称单数 □ ?a-）。

Syncretism occurs when two or more cells of a paradigm with distinct grammatical functions share the same surface form. For instance, ta-, -u realizes multiple feature combinations in Arabic (Figure 1, Table 1). Within highly structured paradigmatic meaning spaces, syncretism directly corresponds to the form-meaning partitions relevant to communicative efficiency; for instance, Zaslavsky et al. (2021b) show that pronoun syncretism patterns are optimized for efficient communication. By definition, syncretism reflects only identity relations between forms (Haspelmath and Sims, 2010). Our proposed model extends beyond syncretism to capture partial overlap in forms, such as the systematic correspondence between the grammatical second person and the prefix ta- in Arabic verbs (Table 1).

融合（syncretism）是指在具有不同语法功能的范式单元之间，共享相同表层形式的现象。例如，ta- 和 -u 在阿拉伯语中实现了多种特征组合（图 1，表 1）。在高度结构化的范式意义空间中，融合直接对应于与交际效率相关的形式-意义划分；例如，Zaslavsky 等人（2021b）表明，代词融合模式是为高效交际而优化的。按定义，融合仅反映形式之间的同一性关系（Haspelmath 和 Sims，2010）。我们提出的模型超越了融合的概念，以捕捉形式的部分重叠，例如阿拉伯语动词中语法第二人称与前缀 ta- 之间的系统性对应关系（表 1）。

2.2 Systematicity in Verb and Pronoun Forms

2.2 动词和代词形式中的系统性

Verbs. Verbal morphology offers a rich testbed to investigate structured forms and meanings. We consider two distinct types of verbal inflection. Concatenative inflection attaches, i.e. concatenates, af-

动词。动词形态为研究结构形式和意义提供了丰富的实验场。我们考虑两种不同的动词变位类型。连缀式变位通过连接（即连缀）来添加，例如：

fixes to a word stem; consider for example the English past tense suffix -ed, as in jump, jumped. Non-concatenative inflection requires changing the word stem; consider for example run, ran. We model Semitic and other Afro-Asiatic verbs, which are characterized by both kinds of morphology. On the non-concatenative side, Semitic roots typically comprise 3 consonants which combine templatically with vowels to produce stems (Huehnergard and Pat-El, 2019b; Weninger, 2011; Lipinski, 1997). These stems are then additionally combined with concatenative morphology in the form of suffixes and sometimes prefixes. For instance, the forms in Figure 1 arise from combining the root verb “to write” KTB with the stem templates and affixes shown in Table 1. We also model verbal morphology in Romance and Germanic languages, where inflection is mainly concatenative.

对词干的修改；例如，英语过去时态后缀 -ed，如 jump（跳）变为 jumped（跳过）。非连缀式屈折变化需要改变词干；例如，run（跑）变为 ran（跑了）。我们建模闪含语和其他非洲-亚非语系动词，这些动词具有这两种形态学特征。在非连缀式方面，闪含语词根通常由三个辅音组成，通过模板与元音结合来产生词干（Huehnergard 和 Pat-El，2019b；Weninger，2011；Lipinski，1997）。这些词干随后还会通过后缀和有时前缀形式的连缀式形态学进行进一步组合。例如，图 1 中的形式是通过将动词词根 “写作”KTB 与表 1 中所示的词干模板和词缀结合而产生的。我们还建模罗曼语系和日耳曼语系中的动词形态学，其中屈折变化主要是连缀式的。

Pronouns. While inflectional morphology reflects systematic form-meaning association by definition (Haspelmath and Sims, 2010, 2), pronouns do not generally display such rich formal structure. Systematicity, however, has also been identified at formal levels below morphology, such as specific sounds (e.g. Blasi et al., 2016; Pimentel et al., 2019; Monaghan et al., 2014). We posit that pronoun forms are also shaped by learnability, and propose modeling their efficiency with an appropriate complexity measure.

代词。虽然屈折形态学按定义反映了系统性的形式-意义关联（Haspelmath 和 Sims，2010，2），但代词通常并不表现出如此丰富的形式结构。然而，系统性也已被识别为低于形态学的形式层面，例如特定的语音（例如 Blasi 等人，2016；Pimentel 等人，2019；Monaghan 等人，2014）。我们提出，代词的形式也受到可学习性的影响，并建议使用适当的复杂度度量来建模其效率。

3 Model

3 模型

We propose an information-theoretic framework to integrate systematicity into efficient communication theory. Our framework builds on a well-established insight: languages evolve under competing pressures for accuracy (enabling precise communication) and simplicity (minimizing cognitive demands). These competing pressures predict that linguistic systems should occupy a particular region of the accuracy-simplicity trade-off space, achieving near-optimal balance between the two.

我们提出一个信息论框架，将系统性整合到高效通信理论中。我们的框架建立在一个已被广泛认可的洞察之上：语言在追求精确性（实现精确交流）和简洁性（最小化认知负担）的竞争压力下演变。这些竞争压力预测，语言系统应该占据精确性与简洁性权衡空间中的特定区域，从而在两者之间实现接近最优的平衡。

The Information Bottleneck (IB) model (Zaslavsky et al., 2018, 2019, 2021b) provides an influential formalization of this trade-off for natural language lexicons. In this model, a speaker samples an object t based on a need distribution $p_{\text{cog}}(t)$ and communicates it by producing a signal w from an encoding distribution $q_{\text{enc}}(w|m_t)$. The listener interprets w using Bayesian inference to construct a probability distribution $q_{\text{dec}}(\hat{m}|w)$ over possible meanings. The encoding process is

信息瓶颈（IB）模型（Zaslavsky 等，2018，2019，2021b）为自然语言词汇库中的这种权衡提供了一个有影响力的正式化描述。在这个模型中，说话者根据需求分布 $p_{\text{cog}}(t)$ 采样一个对象 t，并通过从编码分布 $q_{\text{enc}}(w|m_t)$ 生成一个信号 w 来进行交流。听者使用贝叶斯推理来解释 w，并构建一个可能含义的概率分布 $q_{\text{dec}}(\hat{m}|w)$。编码过程是

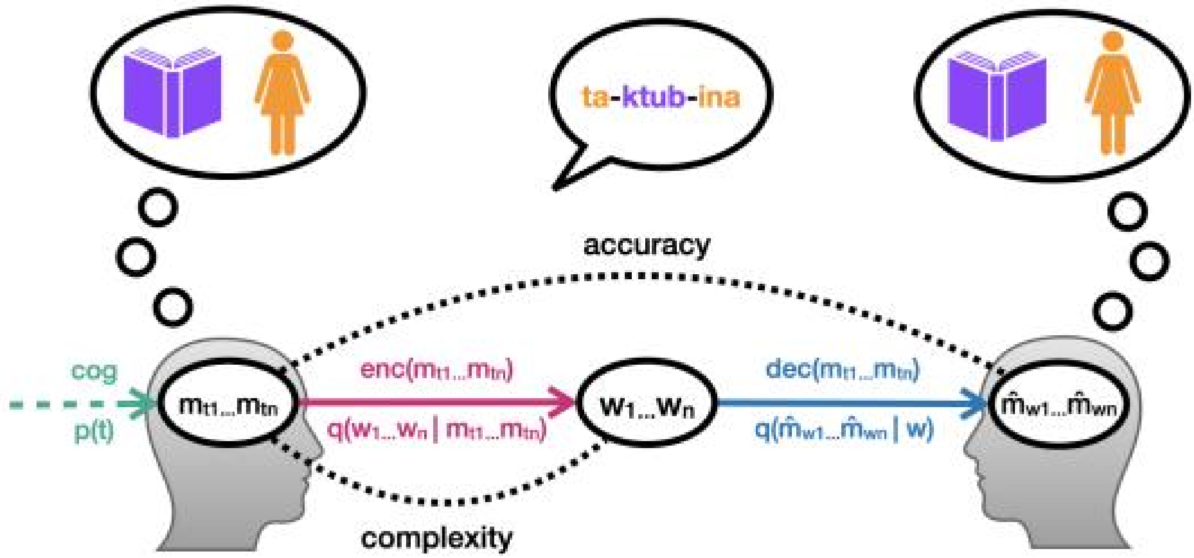

Figure 2: images/40bcc08ca960ea807a4cd1bdb50ed8d7ca9a90f3b16b4bddb12c0ee77d92e2a3.jpg

Figure 2: Communication model, adapted from Zaslavsky et al. (2018, 2021b). Our model encodes the form w as a sequence, and decodes it as an atomic unit.

图 2：通信模型，改编自 Zaslavsky 等人（2018，2021b）。我们的模型将形式 w 编码为一个序列，并将其解码为一个基本单元。

under pressure to minimize complexity, while decoding is pressured to maximize accuracy. The IB framework evaluates this trade-off by comparing attested semantic systems against counterfactual alternatives; if attested systems consistently achieve better complexity-accuracy tradeoffs than counterfactuals, this indicates they have been shaped for communicative efficiency.

在压力下要最小化复杂性，而解码则被压力要求最大化准确性。IB 框架通过将已证实的语义系统与反事实的替代方案进行比较来评估这种权衡；如果已证实的系统在复杂性-准确性权衡方面始终优于反事实方案，这表明它们已经被塑造为更高效的交流方式。

The IB model, however, treats linguistic forms as atomic units with no internal structure. For example, the mappings “3fs □ ta- -u” and “3ms □ ya- -u” (Table 1) could be equivalently represented as “3fs □ X” and “3ms □ Y.” This means the model can detect reduced complexity through syncretism (using identical forms for multiple meanings) but not through systematic form-meaning covariation within forms (e.g., the suffix -u in the third-person singular). Our framework addresses this gap while preserving the IB model’s theoretical foundations and its measure of accuracy (Figure 2).

然而，IB 模型将语言形式视为没有内部结构的原子单位。例如，映射 “3fs □ ta- -u” 和 “3ms □ ya- -u”（表 1）可以等效地表示为 “3fs □ X” 和 “3ms □ Y”。这意味着该模型能够通过合成（使用相同的形式表示多个意义）检测到复杂性的减少，但不能通过形式内部的系统性形式-意义共变（例如第三人称单数的后缀 -u）来做到这一点。我们的框架在保持 IB 模型的理论基础及其准确性度量的同时，弥补了这一不足（图 2）。

Our key contribution is a novel measure of complexity based on the learnability of meaning-to-form mappings. This measure captures systematicity in linguistic forms through a neural network encoder that learns character-level sequences, allowing us to quantify how structural regularities facilitate learning and reduce cognitive complexity.

我们的主要贡献是一种基于意义到形式映射可学习性的新颖复杂度度量方法。该度量方法通过一个学习字符级序列的神经网络编码器，捕捉语言形式中的系统性，使我们能够量化结构规律如何促进学习并降低认

知复杂度。

3.1 Complexity

3.1 复杂性

3.1.1 Our model

3.1.1 我们的模型

We hypothesize that paradigms with more structured mappings will be more learnable. Following the efficient communication literature, we model this with two components: a need distribution $p_{\text{cog}}(t)$, and an encoder $q_{enc}(w|m_t)$.

我们假设具有更结构化映射的范式将更容易学习。根据高效通信文献，我们用两个组件来建模这一现象：一个需求分布 $p_{\text{cog}}(t)$ ，以及一个编码器 $q_{enc}(w|m_t)$ 。

Need distribution. $p_{\text{cog}}(t)$ is a probabilistic weighting over targets t associated with meanings m_t , reflecting communicative need. We assume that more frequent meanings have greater need,

需要分配。$p_{\text{cog}}(t)$ 是对与含义 m_t 相关联的目标 t 的概率加权，反映了交流需求。我们假设更频繁的含义具有更大的需求，

and estimate $p_{\text{cog}}(t)$ using corpus frequency. This weights meanings according to a learner's exposure to different meanings at different frequencies.

并利用语料库频率来估计 $p_{\text{cog}}(t)$。这种方法会根据学习者接触不同意义的频率来对意义进行加权。

Encoder. $q_{enc}(w|m_t)$ maps from a grammatical meaning m_t to a linguistic form w. We implement this using a sequence-to-sequence neural network that takes morphosyntactic features as input (m_t) and generates the corresponding surface form as output (w; Figure 3). [2] Critically, by encoding forms as character sequences rather than atomic units, our encoder can exploit systematic regularities across forms. We expect training to converge more rapidly for more learnable paradigms. We therefore measure complexity in terms of the encoder's cross-entropy decay during learning.

编码器。$q_{enc}(w|m_t)$ 将语法意义 m_t 映射到语言形式 w。我们使用一个序列到序列的神经网络来实现这一功能，该网络以形态句法特征作为输入（m_t），并生成相应的表层形式作为输出（w；图 3）。[2] 关键的是，通过将形式编码为字符序列而不是原子单位，我们的编码器可以利用形式之间的系统性规律。我们期望对于更易学习的词形变化，训练会更快收敛。因此，我们以编码器在学习过程中的交叉熵衰减来衡量复杂度。

Complexity measure. Let $q_{enc}^{(T)}(w|m_t)$ be the encoder after T training epochs. We define the cross-entropy training loss (CETL) as:

复杂度度量。设 $q_{enc}^{(T)}(w|m_t)$ 为 T 个训练周期后的编码器。我们定义交叉熵训练损失（CETL）为：

$$-\frac{\sum_{T=1}^{T_{\max}} \sum_t p_{\text{cog}}(t) \log q_{\text{enc}}^{(T)}(w_t|m_t)}{T_{\max}} \tag{1}$$

Here w_t is the correct form for meaning m_t , and T_{max} is the maximum number of epochs. The core of this equation is $-\log q_{enc}^{(T)}(w_t|m_t)$, which quantifies how well the encoder predicts the form after T training steps. By summing over timesteps, Equation 1 is smaller when training is faster. CETL is therefore information-theoretic, grounded in learnability, and fully general — it can be applied broadly across languages and semantic systems.

这里 w_t 是表示 m_t 的正确形式，而 T_{max} 是最大训练轮数。这个方程的核心是 $-\log q_{enc}^{(T)}(w_t|m_t)$，它量化了编码器在 T 次训练步骤后预测形式的准确性。通过在时间步上求和，当训练速度越快时，方程 1 的值就越小。因此，CETL 是信息论的，基于可学习性，并且具有完全的通用性——它可以广泛应用于各种语言和语义系统中。

Implementation. Our neural network is a LSTM-based sequence-to-sequence (seq2seq) architecture, [3] with two stacked LSTM layers in the encoder and in the decoder. We train with batch size 1 and dropout rate 0.5. Based on preliminary experiments, we set $T_{max} = 50$ epochs [4] and measure total cross-entropy loss weighted by the need distribution. We train ten separate networks for each original paradigm, and five for each counterfactual paradigm, and average over the results. [5]

实现。我们的神经网络是一种基于 LSTM 的序列到序列（seq2seq）架构，[3] 编码器和解码器中均包含两个堆叠的 LSTM 层。我们使用批量大小为 1 和丢弃率 0.5 进行训练。根据初步实验，我们设置 $T_{max} = 50$ 个周期 [4]，并根据需求分布加权计算总交叉熵损失。我们为每个原始范式训练十个独立的网络，为每个反事实范式训练五个，并对结果进行平均。[5]

Figure 3: images/1ab17e6375fad563ea4f7d1b923af44cdef64e80a1560b04d106500759d8b64a.jpg

Figure 3: Example input and output used in training.

图 3：训练中使用的示例输入和输出。

3.1.2 Computational measures of complexity

3.1.2 复杂性的计算度量

Our learnability-based approach connects to several lines of research on linguistic complexity, though CETL uniquely captures systematicity in form-meaning mappings.

我们的基于可学习性方法的思路与多个关于语言复杂性的研究方向相连接，尽管 CETL 在形式-意义映射中独特地捕捉了系统的性。

Learnability-based complexity. Similar neural learnability-based approaches to complexity have been proposed, but none shares our focus on systematicity across forms. Steinert-Threlkeld, Szymanik and colleagues (Steinert-Threlkeld and Szymanik, 2019, 2020; Carcassi et al., 2021) also measure complexity via RNN learning rates within a communicative efficiency framework. Johnson and colleagues (Johnson et al., 2021, 2024) similarly measure morphological complexity via RNN classification accuracy during artificial language learning. These proposed measures, however, rely on classification tasks, and do not attend to forms' internal structure. More recently, Denić and Szymanik (2024) model morphosyntactic complexity in recursive numeral systems, finding an optimal trade-off with lexicon size. This work, like ours, addresses compositional morphological structure; however, it abstracts away from form, instead measuring complexity by average morpheme count. By contrast, CETL detects systematicity across forms (e.g. the shared prefix between "two" and "twenty") without manual annotation. Compared to other learnability-based measures, our approach is more fine-grained, using a continuous information-theoretic measure which is critically sensitive to output forms.

基于可学习性的复杂性。类似基于神经可学习性的复杂性方法已经被提出，但没有一种方法像我们这样关注形式之间的系统性。Steinert-Threlkeld、Szymanik 及其同事（Steinert-Threlkeld 和 Szymanik, 2019, 2020; Carcassi 等, 2021）也在交流效率框架内通过 RNN 学习率来衡量复杂性。Johnson 及其同事（Johnson 等, 2021, 2024）则通过人工语言学习中 RNN 分类准确率来衡量形态复杂性。然而，这些提出的度量方法依赖于分类任务，并未关注形式的内部结构。最近，Denić 和 Szymanik（2024）在递归数制中建模形态句法复杂性，发现与词汇量之间存在最佳权衡。这项工作与我们的研究一样，关注组合形态结构；然而，它忽略了形式本身，而是通过平均词素数量来衡量复杂性。相比之下，CETL 在不进行人工标注的情况下，检测形式之间的系统性（例如，"two" 和 "twenty" 之间的共同前缀）。与其它基于可学习性的度量方法相比，我们的方法更为细致，使用一种连续的信息论度量，对输出形式具有关键的敏感性。

Information-theoretic complexity. Other information-theoretic measures of morphological complexity have been proposed, but most do not fit our specific application. Ackerman and Malouf (2013) define enumerative complexity (e-complexity; essentially the number of formal distinctions) and integrative complexity (i-complexity; the degree to which forms predict each other). They argue that natural languages vary substantially in e-complexity, but tend toward low i-complexity. Variants of this account have been operationalized with seq2seq models (Cotterell et al., 2019; Johnson et al., 2020, 2021). There

信息论复杂度。其他信息论意义上的形态复杂度度量方法也被提出，但大多数并不适用于我们的具体应用场景。Ackerman 和 Malouf（2013）定义了枚举复杂度（e-complexity；本质上是形式上的区分数量）和整合复杂度（i-complexity；形式之间预测程度的高低）。他们认为自然语言在 e-complexity 上存在显著差异，但往往倾向于较低的 i-complexity。这种观点的变体已被通过 seq2seq 模型进行操作化实现（Cotterell 等，2019；Johnson 等，2020, 2021）。There

is a conceptual link to our work, which uses seq2seq learners to model predictive relations between inflected forms. However, i-complexity estimates predictability based on variation across paradigms (e.g., over different inflection classes). Similarly, Wu et al. (2019) quantify the irregularity of inflected forms based on their predictability given the rest of the language. By contrast, we abstract over lexemes and estimate each paradigm's complexity independently from the rest of the language. Rathi et al. (2021) introduce informational fusion, which measures the extent to which a form cannot be predicted by the rest of the paradigm. This measure would be applicable in our domains but has a high computational cost, requiring re-fitting seq2seq models for each form in a paradigm. [6] We prioritize broad coverage, making CETL the most efficient approach.

这是一个与我们工作的概念性联系，我们的工作使用 seq2seq 学习者来建模屈折形式之间的预测关系。然而，i-复杂度估计基于句法类（如不同的屈折类）之间的变化来预测可预测性。同样地，Wu 等人（2019）基于形式在给定语言其余部分的可预测性来量化屈折形式的不规则性。相比之下，我们抽象掉词素，独立于语言其余部分来估计每个句法类的复杂度。Rathi 等人（2021）引入了信息融合，该方法衡量一种形式在句法类其余部分中无法被预测的程度。这一度量方法在我们的领域中是适用的，但计算成本很高，需要为句法类中的每个形式重新拟合 seq2seq 模型。[6] 我们优先考虑广泛的覆盖范围，使 CETL 成为最高效的方法。

IB Complexity. The Information Bottleneck model also uses an information-theoretic formulation of complexity based on need distribution and encoder. For a semantic system with meanings M and forms W, the IB model defines complexity as the mutual information between intended meanings $m_t \in M_t$ and word forms $w \in W$:

信息复杂度。信息瓶颈模型（Information Bottleneck model）也使用基于需求分布和编码器的信息论复杂度定义。对于一个具有意义 M 和形式 W 的语义系统，信息瓶颈模型将复杂度定义为预期意义 $m_t \in M_t$ 和词形 $w \in W$ 之间的互信息：

$$\sum_{\substack{t\in\mathcal{U}\\w\in\mathcal{W}}} p_{\text{cog}}(t) q_{\text{enc}}\left(w \mid m_t\right) \log \frac{q_{\text{enc}}\left(w \mid m_t\right)}{q_{\text{enc}}(w)} \tag{2}$$

where $q_{\text{enc}}(w) = \sum_{t\in\mathcal{U}} p_{\text{cog}}(t) q_{\text{enc}}(w|m_t)$. This computes the bits required for the encoder $q_{\text{enc}}(w|m_t)$ to communicate all meanings, weighted by the need distribution. However, because the IB encoder treats forms as discrete units, this measure of complexity is reduced by syncretism but not by systematic meaning-form covariation. [7] CETL addresses this gap by encoding w as a character sequence, allowing systematic regularities across forms and meanings to reduce complexity through faster learning.

where $q_{\text{enc}}(w) = \sum_{t\in\mathcal{U}} p_{\text{cog}}(t) q_{\text{enc}}(w|m_t)$. 这计算了编码器 $q_{\text{enc}}(w|m_t)$ 为了传达所有含义所需的信息量，其权重由需求分布决定。然而，由于 IB 编码器将形式视为离散单位，因此这种复杂度的衡量标准会因形式的融合而降低，但不会因系统性的意义-形式共变而降低。[7] CETL 通过将 w 编码为字符序列来解决这一差距，从而允许形式和意义之间的系统性规律减少复杂度，通过更快的学习实现这一点。

3.2 Accuracy

3.2 准确性

We adopt the IB model's widely-used, domain-general accuracy measure. Under this framework, a listener uses Bayesian inference to construct a probability distribution $q_{\text{dec}}(\hat{m}|w)$ over possible meanings given the received form w (Figure 2).

我们采用 IB 模型中广泛使用的、适用于各领域的准确性度量方法。在这一框架下，听者利用贝叶斯推断来构建一个概率分布 $q_{\text{dec}}(\hat{m}|w)$，以反映在接收到形式 w 后可能的含义（图 2）。

Accuracy is defined as the similarity between the speaker's intended meaning distribution m_t and the listener's reconstructed distribution $\hat{m}_w$:

准确性被定义为说话者意图含义分布 m_t 与听话者重构分布 $\hat{m}_w$ 之间的相似性：

$$\text{acc} = -\mathbb{E}_{\text{dec}}[D_{KL}[\hat{m}_w || m_t]], \tag{3}$$

where D_{KL} is the Kullback-Leibler (KL) divergence. [8] Following the literature, we assume that speakers always produce the correct form for a target meaning, so m_t is a one-hot vector.

其中 D_{KL} 是 Kullback-Leibler (KL) 散度。[8] 根据文献，我们假设说话者总是为目标含义产生正确的形式，因此 m_t 是一个 one-hot 向量。

To specify $\hat{m}_w$, we require the need distribution $p_{\text{cog}}(t)$ and an underlying representation of the semantic domain, reflecting the listener's knowledge. Following Zaslavsky et al. (2021b), we encode grammatical meanings as categorical features (cf. Regier et al., 2015), and specify their similarity as a weighted Hamming distance $d(u,t)$[9] :

为了指定 $\hat{m}_w$，我们需要需求分布 $p_{\text{cog}}(t)$ 以及语义领域的底层表示，反映听话者的知识。根据 Zaslavsky 等人（2021b）的研究，我们将语法意义编码为类别特征（参见 Regier 等人，2015），并将其相似性指定为加权汉明距离 $d(u,t)$[9]：

$$m_t(u) \propto \exp(-d(u,t)). \tag{4}$$

However, where Zaslavsky et al. (2021b) use binary features, we employ a general categorical encoding scheme. [10] For feature representations u, t, we define $d(u, t)$ as the number of dimensions i where $u_i \neq t_i$. This groups combinatorial meanings with shared features; for example, “first-person singular past” is closer to the meaning “first-person plural past” than “third-person singular future.” Importantly, while this IB accuracy measure posits complex structure in semantic meaning, it still represents the given form w as a discrete atomic unit.

然而，Zaslavsky 等人（2021b）使用的是二元特征，而我们采用了一种通用的类别编码方案。[10] 对于特征表示 u, t，我们定义 $d(u, t)$ 为维度数量 i，其中 $u_i \neq t_i$。这将具有共同特征的组合意义进行分组；例如，“第一人称单数过去式” 与 “第一人称复数过去式” 的含义更为接近，而不是 “第三人称单数将来式”。重要的是，尽管这种 IB 准确性度量假定了语义意义中的复杂结构，它仍然将给定形式 w 视为一个离散的原子单位。

4 Evaluation

4 评估

We formulate two complementary hypotheses about how systematicity affects communicative efficiency in natural language paradigms. To test these hypotheses, we compare attested paradigms against systematically generated counterfactuals. We use this approach to evaluate our model on verb and pronoun data, and compare to the IB framework.

我们提出了两个互补的假设，用以探讨系统性如何影响自然语言范式中的交流效率。为了检验这些假设，我们将已有的范式与系统性生成的反事实范式进行比较。我们采用这种方法来评估我们的模型在动词和代词数据上的表现，并与 IB 框架进行比较。

Hypothesis 1 (Efficiency). Attested paradigms achieve better complexity-accuracy trade-offs than counterfactual alternatives. This follows standard practice in the IB literature (e.g. Kemp and Regier, 2012; Zaslavsky et al., 2021b) and tests whether natural language paradigms are optimized for communicative pressures.

假设 1（效率）。证实的范式在复杂度与准确度之间的权衡上优于反事实的替代方案。这遵循了 IB 文献中的标准做法（例如 Kemp 和 Regier，2012；Zaslavsky 等，2021b），并检验自然语言范式是否针对交流压力进行了优化。

	sg	pl
1	’a-	na-
$2\frac{m}{f}$	ta- ta-	ta–u
ta–u		
$3\frac{m}{f}$	ya- ta- -i	ya–u
ya–u		

	sg	pl
1	’a-	na-
$2\frac{m}{f}$	ta- ta-	ta–u
ta–u		
$3\frac{m}{f}$	ya- ta- -i	ya–u
ya–u		

(a) A structural permutation over the person category (PERS) that increases naturalness: the form ta- now has only value 2 for grammatical person, whereas before it covered values 2, 3.

(a) 一种在人称类别（PERS）上的结构排列，使表达更加自然：现在形式 ta- 在语法人称上的值只有 2，而之前它覆盖的值为 2、3。

	sg	pl
1	’a-	na-
$2\frac{m}{f}$	ta-	ta- -u
ya- -u	ta- -u	
$3\frac{m}{f}$	ya-	ya- -u
ta-	ta- -i	

	sg	pl
1	’a-	na-
$2\frac{m}{f}$	ta-	ta- -u
ya- -u	ta- -u	
$3\frac{m}{f}$	ya-	ya- -u
ta-	ta- -i	

(b) A structural permutation over multiple categories (PERS, NUM) that decreases naturalness: the form ya- -u now has values 2, 3 for grammatical person, and sg, pl for number, whereas before it covered only the feature values 3, pl.

(b) 在多个类别（PERS、NUM）上的结构排列变化，使自然性降低：现在形式 ya–u 的语法人称值为 2、3，数为单数、复数，而以前它只覆盖特征值 3 和复数。

	sg	pl
1	’a-	na-
$2\frac{m}{f}$	ya- -u	ta- -u
ta- -i	ta- -u	
$3\frac{m}{f}$	ya-	ta-
ta-	ta-	

	sg	pl
1	’a-	纳-
$2\frac{m}{f}$	ya- -u	ta- -u
ta- -i	ta- -u	
$3\frac{m}{f}$	ya-	ta-
ta-	ta-	

(c) A form-only permutation over multiple categories (PERS, NUM, GEN) which changes form realizations but does not affect paradigm structure, i.e. the distribution of syncretic forms; compare to Table 2b.

(c) 一种仅在多个类别（PERS、NUM、GEN）上进行形式变化的排列，它改变形式的实现方式，但不影响范式结构，即合成形式的分布；参见表 2b。

Table 2: Example permutations applied to Dialectal Arabic conjugation. Identical (i.e. syncretic) forms whose naturalness is affected by the permutation are underlined. Structural permutations affect syncretic patterns, but form-only permutations do not. Altered forms are highlighted in orange (structural) or cyan (form-only).

表 2：应用于方言阿拉伯语变位的示例排列。那些因排列而自然性受到影响的相同形式（即合成形式）以下划线标出。结构排列会影响合成模式，但仅形式排列则不会。改变的形式以橙色（结构）或青色（仅形式）突出显示。

Hypothesis 2 (Naturalness). Among counterfactual paradigms, those with more natural syncretism patterns (where syncretic forms share similar meanings) are more learnable (Saldana et al., 2022) and thus have lower complexity. This tests whether our model captures not only form-internal, but also paradigm-level structure.

假设 2（自然性）。在反事实范式中，那些具有更多自然融合模式的范式（其中融合形式具有相似的意义）更容易学习（Saldana 等，2022），因此复杂度更低。这测试我们的模型是否不仅捕捉到形式内部的结构，还捕捉到范式层面的结构。

4.1 Generating counterfactual paradigms

4.1 生成反事实范式

We generate counterfactuals by permuting existing paradigms, creating alternatives that vary in two key ways: their syncretism patterns (relevant to systematicity and naturalness) and their specific form-meaning mappings (relevant to systematicity alone). All counterfactuals are compared to their source paradigm as baseline.

我们通过排列现有的范式来生成反事实，创建出在两个关键方面有所不同的替代方案：它们的综合模式（与系统性和自然性相关）以及它们的具体形式-意义映射（仅与系统性相关）。所有反事实都会与它们的源范式进行比较作为基准。

Structural permutations swap the contents of paradigm cells, altering syncretism patterns. For example, we might permute the person feature while holding other distinctions constant, swapping “2sg f □ ta- -i” with “3sg f □ ta-” (Table 2a). We can permute single features (e.g., [2 □ 3]) or multiple features simultaneously (e.g., [2 □ 3] [sg □ pl]; Table 2b). These permutations change which meanings are expressed by syncretic forms, affecting the paradigm’s naturalness score (defined in 4.2).

结构置换会交换范式单元的内容，从而改变合成形式的模式。例如，我们可以在保持其他区别不变的情况下置换人称特征，将 “2sg f □ ta- -i” 与 “3sg f □ ta-” 进行交换（见表 2a）。我们可以置换单个特征（例如 [2 □ 3]），也可以同时置换多个特征（例如 [2 □ 3] [sg □ pl]；见表 2b）。这些置换会改变合成形式所表达的意义，从而影响范式自然度得分（定义见 4.2）。

Form-only permutations swap surface forms while maintaining the original syncretism patterns. For example, we might exchange the form realizing “2sg m” with the form realizing “3sg [m, f],” which retains the original syncretic structure of the paradigm (Table 2c). By definition, these permutations do not affect naturalness scores. Critically, they also cannot affect complexity in the IB model, which treats forms as atomic units. However, they

形式置换（form-only permutations）在保持原有合成模式（syncretism patterns）的前提下，交换表面形式。例如，我们可能会将实现 “2sg m” 的形式与实现 “3sg [m, f]” 的形式进行交换，从而保留该范式原有的合成结构（表 2c）。按照定义，这些置换不会影响自然度评分。关键的是，它们也不会在 IB 模型中影响复杂度，因为该模型将形式视为原子单位。然而，它们

can affect our complexity measure if the permutation disrupts systematic regularities (e.g., breaking a consistent prefix pattern). Form-only permutations thus provide a key test of whether our model captures systematicity beyond syncretism.

可能会对我们复杂度的衡量产生影响，如果排列破坏了系统的规律性（例如，打破了一致的前缀模式）。因此，仅形式上的排列为我们提供了一个关键的测试，以判断我们的模型是否在合成主义之外捕捉到了系统的规律性。

4.2 Measuring naturalness

4.2 测量自然度

Our naturalness hypothesis builds on an insight already present in the IB accuracy measure (3.2): grammatical meanings with shared features are more similar. Systematicity suggests that this principle should extend to forms. In particular, syncretic forms — which share all of their formal features — should also share many aspects of grammatical meaning. For example, the syncretic form na- -u in Table 1 (top two rightmost cells) expresses meanings that all share person = 1. By contrast, an “unnatural” syncretism might use the same form for meanings with no shared features.

我们的自然性假说建立在 IB 准确度度量中已经存在的一个洞察之上（3.2）：具有共享特征的语法意义更为相似。系统性表明，这一原则应扩展到形式上。特别是，合成形式——它们共享所有形式特征——也应该共享许多语法意义的方面。例如，表 1 中（顶部两个最右边的单元格）的合成形式 na- -u 表达了所有共享人称 =1 的意义。相比之下，“不自然” 的合成形式可能会用相同的表达到没有共享特征的意义。

Saldana et al. (2022) develop this intuition as a naturalness gradient in syncretism patterns. Inspired by their analysis, we define the unnaturalness score at the paradigm level:

Saldana 等人（2022）将这一直觉发展为一种自然度梯度，用于同化模式。受他们的分析启发，我们定义了在范式层面的不自然度评分：

$$\text{unnat}_\pi = \sum_{c \in \text{SynClass}_\pi} \sum_{f \in \text{FeatCat}_\pi} (\#c_f - 1) \tag{5}$$

where c_f are all feature values that exist in the syncretism class c for the feature category f. This counts the number of feature values that differ within each syncretic form. A low unnaturalness score indicates natural syncretism patterns where syncretic forms express similar meanings. We compute unnaturalness scores for counterfactual paradigms relative to their attested baselines.

where c_f 是所有存在于合成类 c 的特征类别 f 的特征值。这计算了每个合成形式中特征值的差异数量。较低的不自然性得分表示合成形式表达相似含义的自然合成模式。我们计算反事实范式相对于其已证实基准的不自然性得分。

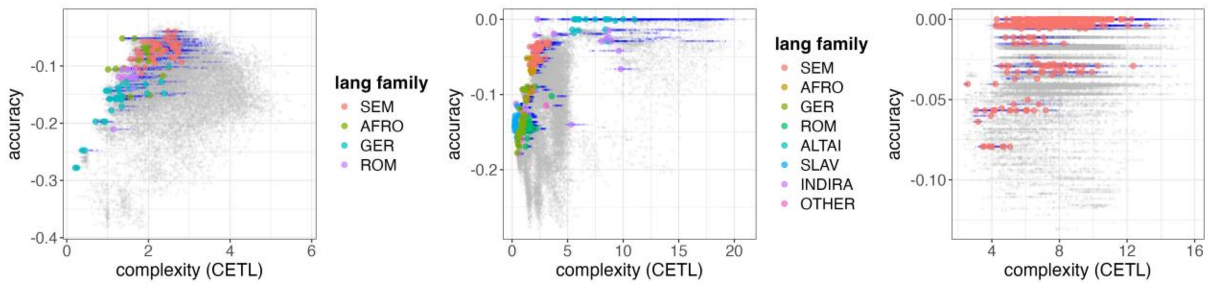

Figure 4: Complexity and accuracy in our model. Across domains, real, attested paradigms (colored, one dot per language) are more efficient than nearly all counterfactual structural (gray) and surface (blue) permutations.

图 4：我们模型中的复杂性与准确性。在不同领域中，真实且有据可查的范式（彩色，每种语言一个点）比几乎所有反事实结构（灰色）和表层（蓝色）排列方式更为高效。

	subsets	#languages	#total permutations	feature categories
PPD		561	93845	PERS, NUM
PRON	PR_SEM	45	16437	
PR_AFRO	19	6677		
PR_GER	30	10649		
PR_ROM	17	3605	PERS, NUM, GEN, CASE	
PR_SLAV	20	6063		
PR_INDOIRAN	18	3616		
PR_ALTAIC	19	3765		
PR_OTHER	9	2264		
VERB	VERB_SEM	56	19700	
VERB_AFRO	13	5773	PERS, NUM, GEN, TEN	
VERB_GER	32	8875		
VERB_ROM	12	2245		

	子集	# 语言	# 所有排列方式	功能分类
PPD		561	93845	PERS, NUM
代词	PR_SEM	45	16437	
非洲人（Afro）	19	6677		
PH0001	30	10649		
PR_ROM	17	3605	PERS, NUM, GEN, CASE	
PR_SLAV	20	6063		
PR_ 印度人	18	3616		
高加索语系	19	3765		
PR_OTHER	9	2264		
动词	动词语义	56	19700	
动词（非洲）	13	5773	PERS, NUM, GEN, TEN	
动词（德语）	32	8875		
动词 _ROM	12	2245		

Table 3: Summary of the datasets. For details of family-specific features within each category, see Appendix A.

表 3：数据集摘要。有关每个类别内特定家族特征的详细信息，请参见附录 A。

4.3 Data

4.3 数据

Table 3 summarizes the languages and features used in our evaluation. Domains 1 and 2b use resources we collected ourselves; see Appendix D for sources and citations.

表 3 总结了我们在评估中使用的语言和特性。领域 1 和 2b 使用了我们自己收集的资源；有关来源和引用，请参见附录 D。

Domain 1: Verbal Morphology (VERB). We test our model on verbal inflection in Semitic, non-Semitic Afro-Asiatic, Germanic, and Romance languages, including both concatenative and non-concatenative morphology. Table 1 illustrates our approach to form and feature representation; see Appendix A for more details.

领域 1：词形变化（VERB）。我们在闪含语、非闪含语的非洲-亚非语系、日耳曼语系和罗曼语系中测试我们的模型，包括连写式和非连写式构词法。表 1 展示了我们对形式和特征表示的方法；更多细节请参见附录 A。

Domain 2a: Pronouns (PPD). To directly compare with Zaslavsky et al. (2021b), we use the Pronoun Paradigms Database (Greenhill, 2025), which contains hundreds of languages across families. We follow their simplified feature scheme (Table 3, top row), which uses only masculine forms and omits dual number.

领域 2a：代词（PPD）。为了与 Zaslavsky 等人（2021b）直接比较，我们使用代词范式数据库（Greenhill, 2025），该数据库包含数百种语言，涵盖多个语系。我们遵循他们简化的特征方案（表 3，顶部行），该方案仅使用阳性形式，并省略了二数。

Domain 2b: Detailed Pronouns (PRON). We also test on full pronoun paradigms for selected language families (Semitic, Germanic), including case and gender. Unlike Domain 2a, the feature set varies by language to capture language-specific distinctions.

领域 2b：详细代词（PRON）。我们还对选定语系（闪含语系、日耳曼语系）的完整代词变体进行测试，包括格和性别。与领域 2a 不同，特征集会根据语言变化，以捕捉语言特有的区别。

Need Distribution. We estimate $p_{cog}(t)$ by combining the need distribution from Zaslavsky et al. (2021b) with additional corpus frequency estimates for gender and dual/plural distinctions, assuming uniform distributions otherwise. Results are robust to this choice (Appendix B).

需要分布。我们通过结合 Zaslavsky 等人（2021b）的需要分布，并加入针对性别和复数/复数区别的额外语料频率估计来估算 $p_{cog}(t)$，假设其他情况下的分布是均匀的。结果对该选择具有鲁棒性（附录 B）。

5 Results

5 结果

We first evaluate whether our model supports the efficiency and naturalness hypotheses (5.1), then compare its performance to the IB approach (5.2). For additional detailed results, see Appendix C.

我们首先评估我们的模型是否支持效率和自然性假设（5.1），然后将其性能与 IB 方法（5.2）进行比较。如需更多详细结果，请参见附录 C。

5.1 Testing efficiency and naturalness

5.1 测试效率和自然性

Efficiency. Figure 4 shows that attested verb and pronoun paradigms are more efficient than nearly all counterfactual permutations: they show lower CETL and higher accuracy, with statistical significance across all domains (details in Appendix C.1). This confirms Hypothesis 1: natural language paradigms occupy a near-optimal region of the complexity-accuracy trade-off space, consistent with communicative efficiency theory.

效率。图 4 显示，经过认证的动词和代词范式比几乎所有反事实排列更加高效：它们表现出更低的 CETL 和更高的准确性，在所有领域中都具有统计学意义（详情见附录 C.1）。这证实了假设 1：自然语言范式位于复杂度-准确性权衡空间中的近最优区域，与交流效率理论一致。

Naturalness. Figure 5 (left panel) shows a positive correlation ($\rho = 0.5745; p < 2.2e - 16$) between CETL and unnaturalness for Afro-Asiatic verbs. Since the unnaturalness scores depend upon paradigm size, we calculate averaged per-language correlations for each domain, finding a lower correlation of 0.36 for PPD, and strong correlations of 0.82 and 0.88 for PRON and VERB respectively. This confirms Hypothesis 2: paradigms with more natural syncretism patterns are more learnable, and thus have lower complexity.

自然性。图 5（左图）显示了 CETL 与非自然性的正相关（$\rho = 0.5745; p < 2.2e - 16$），这一相关性适用于闪含语系的动词。由于非自然性评分取决于词形变化系统的规模，我们为每个领域计算了每种语言的平均相关性，发现 PPD 领域的相关性较低，为 0.36，而 PRON 和 VERB 领域的相关性则较强，分别为 0.82 和 0.88。这证实了假设 2：具有更多自然聚合模式的词形变化系统更容易学习，因此复杂度更低。

Together, these results validate our core hypothesis: systematicity in form-meaning mappings facilitates learning and thereby reduces complexity.

共同的结果验证了我们的核心假设：形式与意义之间的系统性映射有助于学习，从而减少复杂性。

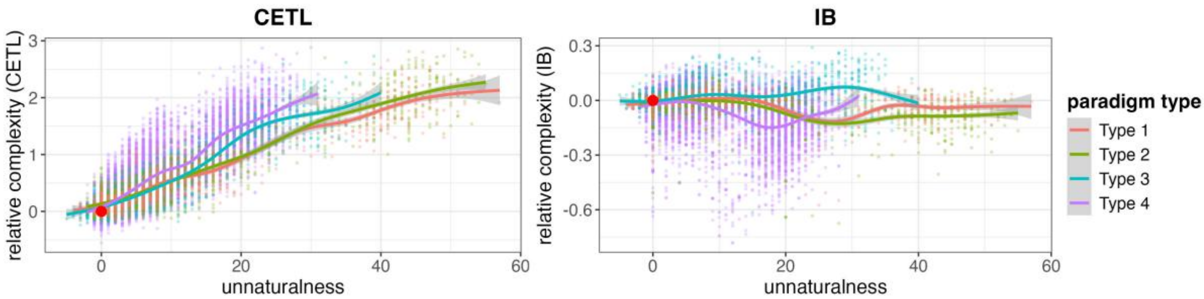

Figure 5: Complexity plotted against unnaturalness for Afro-Asiatic verbs by model, CETL (left) and IB (right). We group results by paradigm type (reflecting different grammatical categories; cf. App. A.2). CETL positively correlates with unnaturalness, meaning more natural paradigms have lower complexity, while IB shows no correlation.

图 5：按模型（CETL（左）和 IB（右））对非洲-亚洲语系动词的复杂性与不自然性进行绘制。我们按词类类型对结果进行分组（反映不同的语法类别；参见附录 A.2）。CETL 与不自然性呈正相关，意味着更自然的词类具有更低的复杂性，而 IB 则没有显示出相关性。

	C_M (%)		I_M (%)		$Perf_M$		support
CETL	IB	CETL	IB	CETL	IB		
PPD	72.37	4.82	2.19	0.73	70.19	4.10	44112
PRON	90.12	48.71	1.48	4.53	88.64	44.18	38623
VERB	76.32	41.03	3.77	4.08	72.55	36.95	32825

	C_M (%)		I_M (%)		$Perf_M$		支持
CETL	IB	CETL	IB	CETL	国际货币基金组织（IMF）		
PPD	72.37	4.82	2.19	0.73	70.19	4.10	44112
代词	90.12	48.71	1.48	4.53	88.64	44.18	38623
动词	76.32	41.03	3.77	4.08	72.55	36.95	32825

(a) Structural Permutations

(a) 结构排列

	C_M (%)		I_M (%)		$Perf_M$		support
CETL	IB	CETL	IB	CETL	IB		
PPD	65.81	0.00	34.19	100	31.63	-100	51061
PRON	89.21	0.00	10.79	100	78.42	-100	10697
VERB	78.23	0.00	21.77	100	56.47	-100	3781

	C_M (%)		I_M (%)		$Perf_M$		支持
CETL	IB	CETL	IB	CETL	IB		
PPD	65.81	0.00	34.19	100	31.63	-100	51061
代词	89.21	0.00	10.79	100	78.42	-100	10697
动词	78.23	0.00	21.77	100	56.47	-100	3781

(b) Form-only Permutations
Table 4: Performance for structural (left) and surface permutations (right). CETL outperforms IB across the board.

(b) 仅形式的排列
表 4：结构排列（左侧）和表面排列（右侧）的性能表现。CETL 在各个方面都优于 IB。

5.2 Comparison to the IB Model

5.2 与 IB 模型的比较

Our CETL measure is sensitive to surface form structure, which we expect to increase discriminative capacity relative to the IB model's complexity measure. Specifically, we expect CETL to correctly identify attested paradigms as more efficient than counterfactual permutations, and show a stronger correlation than IB to paradigm naturalness.

我们的 CETL 度量对表层形式结构敏感，我们预计它相对于 IB 模型的复杂度度量而言，能够提高判别能力。具体而言，我们预计 CETL 能够正确识别已证实的词形变化模式比假设的排列组合更为高效，并且与 IB 相比，与词形变化的自然性具有更强的相关性。

Evaluation. We measure each model's ability to correctly identify attested paradigms as more efficient than counterfactuals. For each counterfactual permutation, we classify each estimate as either Correct(C): worse than the attested paradigm in both accuracy and complexity, or worse in one measure while equal in the other — or Incorrect(I): better than or equal to the attested paradigm in both measures. We define the performance of a model M as $\text{Perf}_M = C_M - I_M$ (i.e., the difference between correct and incorrect identifications). We consider CETL to outperform the IB model when it performs better by at least 5% of the permutations for a given language.

评估。我们衡量每个模型正确识别认证范式的能力是否比反事实范式更高效。对于每个反事实排列，我们将每个估计值分类为正确（C）：在准确性和复杂性两个指标上都比认证范式差，或者在一个指标上更差而另一个指标相等；或者错误（I）：在两个指标上都优于或等于认证范式。我们将模型 M 的性能定义为 $\text{Perf}_M = C_M - I_M$（即正确识别与错误识别之间的差异）。我们认为 CETL 在某种语言上优于 IB 模型，当其性能至少比 IB 模型好 5% 的排列时。

Results. Table 9b shows that CETL correctly identifies 65% to nearly 90% of form-only permutations as less efficient than attested paradigms, while the IB model cannot distinguish these permutations at all (by design, since they have identical syncretism patterns). This demonstrates that CETL captures systematicity beyond mere syncretism. Table 9a shows that CETL consistently outperforms

结果。表 9b 显示，CETL 能够正确识别出形式排列中 65% 至近 90% 的排列方式比已证实的范式效率更低，而 IB 模型完全无法区分这些排列方式（按设计，因为它们具有相同的合成模式）。这表明 CETL 捕捉到了超越单纯合成的系统性。表 9a 显示，CETL 始终优于其他模型。

the IB model in identifying attested paradigms as more efficient than structural permutations. Finally, Figure 5 shows that CETL (left) strongly correlates with naturalness, with correlations exceeding 0.8 for PRON and VERB, while the IB model (right) shows no correlation for either domain. This confirms that CETL successfully captures fine-grained regularities in syncretism patterns that the IB model cannot detect. We conclude that our learnability-based complexity measure better captures systematic form-meaning mappings, enabling better discrimination between attested and counterfactual paradigms.

在识别证实的范式方面，IB 模型比结构排列更有效。最后，图 5 显示，CETL（左侧）与自然性有很强的相关性，对于代词（PRON）和动词（VERB）的相关性超过 0.8，而 IB 模型（右侧）在两个领域都没有相关性。这证实了 CETL 成功捕捉到了 IB 模型无法检测到的合成模式中的细粒度规律。我们得出结论，基于可学习性的复杂性度量更好地捕捉了系统性的形式-意义映射，从而能够更好地区分证实的范式和反事实范式。

6 Conclusion

6 结论

Languages vary widely in form-meaning mappings. We propose a unified framework that integrates systematicity between meaning and form into efficient communication theory, and evaluate our model on hundreds of languages from diverse families. Our results show that: (1) attested paradigms are more efficient than counterfactuals; (2) paradigms with more natural syncretism patterns are more learnable; and (3) our model outperforms the IB approach in discriminating attested from counterfactual paradigms. This work connects efficient communication theory with language evolution research on systematicity, showing how communicative pressures shape language.

语言在形式与意义的映射上差异很大。我们提出一个统一的框架，将意义与形式之间的系统性整合到高效沟通理论中，并在来自不同语系的数百种语言上评估我们的模型。我们的结果表明：(1) 已证实的词形变化模式比假想的模式更高效；(2) 具有更多自然融合模式的词形变化更易于学习；(3) 我们的模型在区分已证实与假想词形变化模式方面优于 IB 方法。这项工作将高效沟通理论与关于系统性的语言演化研究联系起来，展示了沟通压力如何塑造语言。

Limitations

限制

Our analysis focuses exclusively on discrete, paradigmatically structured domains—specifically verbal inflection and pronouns. The applicability of CETL to other semantic domains remains uncertain. In particular, continuous domains like color present a fundamental challenge: they lack the discrete categorical structure required by our sequence-to-sequence architecture.

我们的分析仅专注于具有范式结构的离散领域——特别是动词变位和代词。CETL 在其他语义领域的适用性尚不确定。特别是像颜色这样的连续领域提出了根本性的挑战：它们缺乏我们序列到序列架构所需的离散分类结构。

While CETL captures fine-grained systematicity that the IB measure misses, we have not established whether it is universally preferable or whether trade-offs exist. For instance, we do not replicate the fine-grained feature weight optimization of Za-slavsky et al. (2021b), leaving open whether CETL supports such analyses as effectively. The two measures may prove complementary, with different applications suited to each.

虽然 CETL 能够捕捉到 IB 度量所忽略的细粒度系统性，但我们尚未确定它是否在所有情况下都是更优的，或者是否存在权衡。例如，我们没有复制 Za-slavsky 等人（2021b）的细粒度特征权重优化，因此无法确定 CETL 是否能同样有效地支持此类分析。这两个度量可能具有互补性，各自适用于不同的应用场景。

Lastly, unlike some prior work in the efficient communication literature, we do not provide an explicit Pareto frontier of maximally efficient paradigms. Doing so would require specifying all possible counterfactual paradigms, including all forms compatible with the phonology and phonotactics of each language. Our permutation-based approach generates structured counterfactuals but does not exhaustively sample the space of possibilities, limiting our ability to make strong optimality claims about attested systems.

最后，与一些先前在高效通信文献中的工作不同，我们并未提供一个显式的帕累托前沿，以展示最大效率范式的集合。要做到这一点，需要指定所有可能的反事实范式，包括所有与每种语言的音系和音系规则兼容的形式。我们的基于排列的方法可以生成结构化的反事实范式，但并未穷尽所有可能性的空间，这限制了我们对已记录系统做出强最优性声明的能力。

Ethics Statement

伦理声明

This paper concerns foundational research on the structure of language. We do not foresee immediate ethical implications.

本文涉及语言结构的基础研究。我们不预见到近期的伦理影响。

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